

Problem 1 10 marks

A company has been assigned the following IP address range: 192.168.1.0/24. They have the following network requirements: Network A: 30 hosts, Network B: 10 hosts, and Network C: 2 hosts

1. Design a subnetting scheme using VLSM to accommodate these network requirements.
2. Determine the subnet masks for each network.
3. Calculate the usable host range for each subnet.

0-127	A
128-191	B
192-223	C
224-239	D
240-255	E

Problem 2 10 marks

A user is unable to connect to a network resource with the IP address 192.168.1.100. The user's device shows the correct IP address configuration (192.168.1.101) and default gateway (192.168.1.1). Network connectivity to other resources on the same subnet is successful.

1. What are the possible causes of this IP address mismatch issue?
2. How can you troubleshoot this problem to determine the root cause?
3. What steps can be taken to resolve the issue once the cause is identified?

Tutorial: 02

Course: ICT-3107 (Information & Data Security)

Full Marks: 20

Time: 40 Minutes

of Three

Answer any TWO of the following questions.

- 1a) Define cryptography. List some widely used cryptosystem. (5)
- b) State four security needs provided by cryptography. (5)
- 2a) Give a general idea behind symmetric-key cryptography. (5)
- b) How does asymmetric-key cryptography provide the message confidentiality _____? (5)
- 3a) What are the merits and demerits of asymmetric-key cryptography over symmetric-key cryptography? _____ (5)
- b) Three pass protocol can be used to send sensitive information across an insecure network. Give a postal analogy. (5)